Mock 11 + Mock 12: Wrong-Answer Notes

CFA Level II · 16 wrong answers + 2 added questions, one per page

72 / 88	40 / 44	32 / 44
Combined (81.8%)	Mock 11 (90.9%)	Mock 12 (72.7%)

Score by topic (Mock 11 + 12 combined)

Topic	Score	%
Derivatives	4/8	50%
Financial Statement Analysis	8/12	67%
Fixed Income	9/12	75%
Alternative Investments	3/4	75%
Corporate Issuers	7/8	88%
Quantitative Methods	7/8	88%
Equity Valuation	11/12	92%
Ethical and Professional Standards	11/12	92%
Economics	8/8	100%
Portfolio Management	4/4	100%

Questions at a glance

Q	Topic	You	Ans
Mock 11 Q23	Equity	A	B
Mock 11 Q30	Deriv	C	B
Mock 11 Q31	Deriv	B	A
Mock 11 Q39	FSA	A	B
Mock 12 Q2	Deriv	B	A
Mock 12 Q3	Deriv	A	B
Mock 12 Q6	FSA	A	B
Mock 12 Q7	FSA	C	A
Mock 12 Q9	Alt	C	A
Mock 12 Q16	FSA	C	A
Mock 12 Q18	Corp	B	A
Mock 12 Q26	Ethics	A	C
Mock 12 Q29	FI	B	A
Mock 12 Q37	FI	C	A
Mock 12 Q39	FI	B	C
Mock 12 Q42	Quant	B	C

Mock 11 Q23 **PEG ratio: forward P/E (price ÷ NEXT-four-quarter EPS) ÷ growth rate in %**

You **A** · Answer **B**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Alves directs Ramos to use a valuation metric that would allow for a meaningful ranking of relative value of the three companies' shares. Exhibit 1 provides selected financial information for the three companies.

Exhibit 1: Selected Financial Information for Norte, Costa, and Valle (Mexican Peso, MXN)

	Norte	Costa	Valle
Stock's current price	18.60	64.80	118.30
Diluted EPS (last four quarters)	0.95	3.35	-0.12
Diluted EPS (next four quarters)	1.05	2.70	2.35

Exhibit 2: Earnings Growth Rate Estimates over Five Years

Company	Earnings Growth Rate Estimate (%)
Norte	15.40
Costa	24.50
Valle	28.60

Question 23. Based on Exhibit 1 and Exhibit 2, which company's shares are the most attractively priced based on the five-year forward P/E-to-growth (PEG) ratio?

OPTIONS

A	Norte	✗ Your pick	Your pick: Norte has a low forward P/E (17.7) but also the lowest growth (15.40%), so its PEG (1.15) is above Costa's.
B	Costa	✓ Correct	Correct: Costa has the lowest forward PEG (0.98).
C	Valle		Valle has the highest forward PEG (1.76): a high forward P/E (50.3) is not paid for by its growth, even though its growth is the highest.

Trigger "five-year forward P/E-to-growth (PEG) ratio"
"Diluted EPS (next four quarters)"
"Earnings Growth Rate Estimate (%)"

Rule next time: Forward PEG = (price ÷ EPS of the next four quarters) ÷ growth rate (as a number, 15.40 not 0.154). The LOWEST PEG is the cheapest per unit of growth.

FORMULA / RULE

Forward P/E = price ÷ forward EPS
PEG = forward P/E ÷ (growth rate × 100)
Lowest PEG = most attractively priced

WHY THE ANSWER IS B

1. Norte: 18.60 ÷ 1.05 = 17.71; ÷ 15.40 = 1.15.
2. Costa: 64.80 ÷ 2.70 = 24.00; ÷ 24.50 = 0.98.
3. Valle: 118.30 ÷ 2.35 = 50.34; ÷ 28.60 = 1.76: **B**.

Apply to other questions: "Forward" → next-four-quarter EPS; compute all three companies before choosing; a low P/E alone is not enough, divide by the growth.

Mock 11 Q30 **FRA value after initiation: new FRA rate from MRRs, then discount to TODAY over the days to the END of the underlying period**

You **C** · Answer **B**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

One month ago (30 days), Wells entered a pay-fixed 3 × 6 forward rate agreement (FRA) at a rate of 2.40% with a notional amount of US$5,000,000. Now, 30 days after entering the FRA, the two-month MRR is 1.5% and the five-month MRR is 2.5%.

Question 30. The current value of Wells's FRA is closest to:

OPTIONS

A	US$9,485.		Leaves the settlement undiscounted (9,485).
B	US$9,387.	✓ Correct	Correct: discounted over 150 days = US$9,387.
C	US$9,426.	✗ Your pick	Your pick: discounts over 90 days (the length of the underlying period) instead of 150 days (time from today to its end): US$9,426.

Trigger “pay-fixed 3 × 6 forward rate agreement (FRA) at a rate of 2.40%”
“30 days after entering the FRA”
“the two-month MRR is 1.5% and the five-month MRR is 2.5%”

Rule next time: After 30 days the FRA has 2 months left and the underlying 3-month period ends in 5 months (150 days). Find the new 2 × 5 rate, take the difference with the contract rate, and discount over 150 days (the end of the underlying period), not 90.

FORMULA / RULE

New FRA rate = [(1 + L_long × t_long/360) ÷ (1 + L_short × t_short/360) − 1] × 360/(t_long − t_short)
Value (pay-fixed) = (new rate − contract rate) × (days/360) × notional ÷ (1 + L_long × t_long/360)

WHY THE ANSWER IS B

1. New rate = [(1 + 0.025 × 150/360) ÷ (1 + 0.015 × 60/360) − 1] × 360/90 = 3.159%.
2. Difference: (3.159% − 2.40%) × 90/360 × 5,000,000 = 9,485.
3. Discount: 9,485 ÷ (1 + 0.025 × 150/360) = US$9,387: **B**.

Apply to other questions: FRA value: the discount period is from today to the END of the underlying period, i.e. the longer MRR (here 150 days), never the length of the period itself.

Mock 11 Q31 **Currency swap fixed rate: use the PV factors of the leg you PAY; sum of all factors in the denominator**

You **B** · Answer **A**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Rao, who is based in Hong Kong SAR, requires a one-year bridge loan in euros to fund operations in Germany. He wants to fund this loan at a competitive rate. Fischer advises Rao to borrow HK dollars domestically and then enter into a one-year foreign currency swap to swap into euros. Exhibit 1 below provides Hong Kong and euro spot interest rates and present value factors.

Exhibit 1: Hong Kong and Euro Spot Interest Rates

Days to Maturity	HK Dollars Spot Interest Rates (%)	HK Dollars Present Value Factors	Euro Spot Interest Rates (%)	Euro Present Value Factors
90	0.620	0.9985	0.355	0.9991
180	0.780	0.9961	0.405	0.9980
270	0.860	0.9936	0.440	0.9967
360	0.950	0.9906	0.470	0.9953

Question 31. Based on the information in Exhibit 1, the annual fixed swap rate Rao would pay is closest to:

OPTIONS

A	0.47%.	✓ Correct	Correct: the euro fixed rate Rao pays, 0.47%.
B	0.95%.	✗ Your pick	Your pick: 0.95% is the HK dollar fixed rate, which Rao RECEIVES.
C	1.89%.		Divides by the last factor instead of the sum of the factors (1.89%).

Trigger “requires a one-year bridge loan in euros”
“borrow HK dollars domestically and then enter into a one-year foreign currency swap to swap into euros”
“Euro Present Value Factors”

Rule next time: Rao needs euros, so in the swap he receives HK dollars (to pay his HK dollar loan) and PAYS euros: use the EURO factors. Fixed rate per period = (1 − last PV factor) ÷ sum of the PV factors; multiply by 4 for the annual rate.

FORMULA / RULE

Fixed rate per period = (1 − PV_n) ÷ Σ PV_i
Annual rate = per-period rate × periods per year (quarterly: × 4)

WHY THE ANSWER IS A

1. Euro leg: (1 − 0.9953) ÷ (0.9991 + 0.9980 + 0.9967 + 0.9953) = 0.1178%.
2. × 4 = 0.47%: **A**.
3. HK dollar leg: (1 − 0.9906) ÷ 3.9788 × 4 = 0.95% (the leg Rao receives).

Apply to other questions: Currency swap: decide which currency you PAY first, then take that currency's factors. The borrower ends up paying the currency he needs.

Mock 11 Q39 **Equity method carrying value: cost + share of net income − share of dividends − amortisation of the fair value excess**

You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Investment in Calder Co.

On 1 January 2016, Veridian acquired a 30% equity interest in Calder Co. for $620 million. Exhibit 2 presents selected financial information for Calder on the acquisition date. Thornton notes that the plant and equipment are depreciated on a straight-line basis and have 10 years of remaining life.

Exhibit 2: Selected Financial Data for Calder Co., 1 January 2016 (Acquisition Date) ($ millions)

	Book Value	Fair Value
Current assets	300	300
Plant and equipment	3,100	3,400
Total assets	3,400	3,700
Liabilities	1,900	1,900
Net assets	1,500	1,800

Thornton notes that, for fiscal year 2016, Calder reported net income of $400 million and paid dividends of $240 million.

Question 39. Using only the information from Exhibit 2 and the 2016 results, the carrying value of Veridian's investment in Calder at the end of 2016 is closest to:

OPTIONS

A	$668 million.	✗ Your pick	Your pick: leaves out the amortisation of the plant excess (668).
B	$659 million.	✓ Correct	Correct: 620 + 120 − 72 − 9 = $659 million.
C	$731 million.		Does not deduct the dividends received (731).

Trigger “depreciated on a straight-line basis and have 10 years of remaining life”
“Plant and equipment | 3,100 | 3,400”
“reported net income of $400 million and paid dividends of $240 million”

Rule next time: Fair value above book value on depreciable plant creates an excess that must be amortised: 30% × (3,400 − 3,100) ÷ 10 years = 9 per year. Carrying value = cost + 30% × net income − 30% × dividends − 9.

FORMULA / RULE

> Investment = cost + % × net income − % × dividends − amortisation
> Amortisation = % × (FV − BV) ÷ remaining life (for the part of the excess on depreciable assets)

WHY THE ANSWER IS B

1. Share of net income: 0.30 × 400 = 120; dividends: 0.30 × 240 = 72.
2. Amortisation: 0.30 × (3,400 − 3,100) ÷ 10 = 9.
3. 620 + 120 − 72 − 9 = **659**: **B**.

Apply to other questions: Equity method: dividends reduce the investment; and check whether FV differs from BV on depreciable assets (amortise) or on land (no amortisation).

Mock 12 Q1 Expectations approach to option value: RISK-NEUTRAL probabilities, discount at the risk-free rate

You **C** · Answer **C**

Deriv

Answered correctly; added at your request

VIGNETTE (RELEVANT PART, UNEDITED)

Ortiz comments: "I think six-month call options on the six-month forward rate would probably be the cheapest solution. The price of the European-style option can be evaluated as the present value of the expected terminal option payoffs, using the true probabilities of the underlying rate moves and discounting at the risk-free periodic rate."

Question 1. Ortiz's understanding of the valuation of the European-style six-month call option is most likely:

OPTIONS

A	correct with respect to the payoffs but incorrect about the discount rate.		Wrong: the discount rate is not the error.
B	correct with respect to the payoffs and the discount rate.		Wrong: the payoffs use the wrong probabilities (true instead of risk-neutral).
C	incorrect with respect to the payoffs but correct about the discount rate.	✓ Correct	Correct: payoffs wrong (true probabilities), discount rate right (risk-free).

Trigger “using the true probabilities of the underlying rate moves”
“discounting at the risk-free periodic rate”

Rule next time: Option value = present value of the expected payoffs computed with RISK-NEUTRAL probabilities, discounted at the risk-free rate. 'True' (real-world) probabilities are the error; the discount rate is right.

FORMULA / RULE

> Option value = PV at the risk-free rate of E*[payoff], with E* under risk-neutral probabilities

WHY THE ANSWER IS C

1. Risk-neutral probabilities make the expected value of the underlying grow at the risk-free rate, so the risk-free rate is also the discount rate.
2. Real-world probabilities would need a risk-adjusted discount rate: mixing them with the risk-free rate is the mistake Ortiz makes: **C**.

Apply to other questions: Spot the two parts of every valuation statement (probabilities, discount rate) and test each one separately.

Mock 12 Q2 Swaption hedge: payer swaption = swap component − bond component (receiver = bond − swap)

You **B** · Answer **A**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Halvorsen states: "Ridgemont University plans a laboratory renovation for its science center and has engaged DRA to implement options strategies in order to manage the risk of rising interest rates. The renovation is to be completed in 12 months, in time for the start of the school year. To minimize disruption to its academic schedule, however, Ridgemont will not begin the work until six months from now. State funding will not be received until the beginning of the next school year, so a six-month variable interest rate loan will finance the renovation."

Halvorsen comments: "We can also consider options on swaps, which the Black model views as having a bond component and a swap component. The swaption, used to hedge against rising interest rates, can be evaluated as the bond component minus the swap component."

Question 2. Is Halvorsen's description of the swaption used for the hedge most likely correct?

OPTIONS

A	No, because a swaption used to hedge against rising rates would be evaluated as the swap component minus the bond component.	✓ Correct	Correct: payer swaption is swap component minus bond component, so Halvorsen reversed the components.
B	No, because the swaption used to hedge against rising rates is a receiver swaption.	✗ Your pick	Your pick: a swaption that hedges against rising rates is a PAYER swaption, not a receiver swaption.
C	Yes		Wrong: the description is not correct.

Trigger “manage the risk of rising interest rates”
“The swaption, used to hedge against rising interest rates”
“bond component minus the swap component”

Rule next time: Hedging RISING rates = PAYER swaption (right to pay fixed), valued in the Black model as swap component − bond component. A receiver swaption (hedges FALLING rates) = bond component − swap component.

FORMULA / RULE

> Payer swaption value (Black) = swap component − bond component
> Receiver swaption value (Black) = bond component − swap component

WHY THE ANSWER IS A

1. Rising rates hurt a floating-rate borrower: the payer swaption locks in a fixed rate to pay.
2. In the Black model a payer swaption is a call on the swap rate, a receiver swaption is a put on the swap rate.
3. Halvorsen reversed the components: **A**.

Apply to other questions: Rates up → payer (pay fixed, call on rate); rates down → receiver (receive fixed, put on rate). Then match the component order.

Mock 12 Q3 **Delta hedge: number of options = shares ÷ option delta (absolute value), never shares × delta**

You **A** · Answer **B**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Ridgemont has an endowment that helps support the university financially. The university has learned the endowment will be receiving a gift of 150,000 shares of Pelham Group (PG) stock in one month. DRA recommends the use of a one-month options position to hedge against a material price decline on the stock during this period. Exhibit 1 lists the relevant PG stock and option characteristics.

Exhibit 1: PG Stock and Option Information

Item	Value
Stock price	$64
Call and put options strike price	$62
Delta call	0.6217
Delta put	–0.3783

Question 3. Assuming one option per share, an appropriate delta hedge for the PG stock would most likely be to:

OPTIONS

A	buy 56,745 puts.	✗ Your pick	Your pick: multiplies shares by the put delta (150,000 × 0.3783 = 56,745); the number of puts is shares ÷ |put delta|.
B	sell 241,274 calls.	✓ Correct	Correct: sell 150,000 ÷ 0.6217 = 241,274 calls.
C	sell 93,255 calls.		Multiplies by the call delta (150,000 × 0.6217 = 93,255).

Trigger “150,000 shares”
“hedge against a material price decline”
“Delta call | 0.6217”

Rule next time: Long 150,000 shares has a delta of +150,000. To make the position delta-neutral SELL calls: 150,000 ÷ 0.6217 = 241,274. (The put route would be BUY 150,000 ÷ 0.3783 = 396,511 puts.) Divide by the delta, do not multiply.

FORMULA / RULE

Options needed = shares ÷ |delta of the option|
Long stock: sell calls or buy puts; delta of the options must offset the shares

WHY THE ANSWER IS B

1. Call route: 150,000 ÷ 0.6217 = 241,274 calls to SELL.
2. Put route: 150,000 ÷ 0.3783 = 396,511 puts to BUY.
3. Answer B.

Apply to other questions: Delta hedge: each option offsets only 'delta' of one share, so you need MORE options than shares.

Mock 12 Q6 **Current rate method: ALL income statement items at the average rate (the temporal method uses the inventory-purchase rate for COGS)**

You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Arica Corp. and Bosque Corp. are two wholly owned subsidiaries of Brisa that are located in the US. There are no intercompany transactions. In both subsidiaries, the only non-monetary assets are inventory and fixed assets, and all liabilities are monetary.

Brisa translates Arica's financial statements using the current rate method. In the most recent year, Arica's revenue was USD 90 million and cost of sales was USD 52 million. The average exchange rate for inventory purchases was 930.00 CLP/USD.

Exhibit 1: Exchange Rates

Date	Exchange Rate CLP/USD
15 November	905.00
31 December	880.00
Average for the year	915.00

Question 6. Arica's contribution to Brisa's consolidated gross profit (in CLP millions) in the most recent year is closest to:

OPTIONS

A	33,990.	✗ Your pick	Your pick: cost of sales at 930 (temporal method) while the method for Arica is the current rate.
B	34,770.	✓ Correct	Correct: (90 − 52) × 915.00 = 34,770.
C	33,440.		Gross profit at the year-end rate (880): balance sheet rate, not an income statement rate.

Trigger “Brisa translates Arica's financial statements using the current rate method”
“Average for the year | 915.00”

Rule next time: Current rate method: revenues AND expenses (including cost of sales) at the average rate → gross profit = (90 − 52) × 915.00. The inventory-purchase rate (930) belongs to the temporal method (Bosque).

FORMULA / RULE

> Current rate: revenues and expenses at the average rate; assets and liabilities at the closing rate
> Temporal: COGS and depreciation at the historical rate

WHY THE ANSWER IS B

1. 90 × 915.00 = 82,350; 52 × 915.00 = 47,580; gross profit 34,770.
2. Temporal would give 90 × 915.00 − 52 × 930.00 = 33,990 (the trap).
3. Year-end rate: 38 × 880.00 = 33,440 (the other trap): **B**.

Apply to other questions: First find the METHOD named for each subsidiary, then apply its rates; do not mix the two methods' rates.

Mock 12 Q7 **Current rate vs temporal method: where the exchange effect goes and which exposure it measures**

You **C** · Answer **A**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Arica Corp. and Bosque Corp. are two wholly owned subsidiaries of Brisa that are located in the US. There are no intercompany transactions. In both subsidiaries, the only non-monetary assets are inventory and fixed assets, and all liabilities are monetary.

Brisa uses the temporal method to translate Bosque's financial statements. Soto evaluates the effect on Brisa's total shareholders' equity if Bosque's financial statements had been translated using the same method as Arica.

Exhibit 1: Exchange Rates

Date	Exchange Rate CLP/USD
15 November	905.00
31 December	880.00
Average for the year	915.00

Question 7. If Bosque's results had been translated using the current rate method, its total shareholders' equity would have been:

OPTIONS

A	lower.	✓ Correct	Correct: current rate puts a negative translation adjustment in equity, so equity is lower.
B	higher.		Higher only if the USD had risen against the CLP.
C	the same.	✗ Your pick	Your pick: the two methods are not equivalent; they differ in the exposure and in where the effect goes (net income vs OCI).

Trigger “the only non-monetary assets are inventory and fixed assets, and all liabilities are monetary”
“translated using the same method as Arica”
“905.00” → “880.00”

Rule next time: Temporal: exposure is net MONETARY items (net monetary liability + falling foreign currency = gain in net income). Current rate: exposure is ALL net assets (falling foreign currency = negative CTA in equity). With the USD falling against the CLP, equity is lower under the current rate method.

FORMULA / RULE

> Temporal: monetary items at the closing rate, non-monetary at historical rates; gain/loss in net income
> Current rate: all assets and liabilities at the closing rate; CTA in OCI (equity)

WHY THE ANSWER IS A

1. USD fell from 905 to 880 CLP/USD: the subsidiary's net assets are worth less in CLP.
2. Current rate: all net assets are re-translated lower → negative CTA, equity lower.
3. Temporal: net monetary liability gains when the foreign currency falls → gain in net income: **A**.

Apply to other questions: Use your translation table: foreign currency falls + net asset exposure → loss/negative CTA; net liability exposure → gain.

Mock 12 Q9 **Backwardation = near futures price ABOVE the farther prices (contango = near below farther)**

You **C** · Answer **A**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Exhibit 1: Selected Commodity Futures Data*

Month	Gold Price	Coffee Price	Heating Oil Price
July	1,274.8	0.9850	2.4105
September	1,274.8	1.0030	2.3552
December	1,274.8	1.0290	2.1980

Question 9. Which futures market in Exhibit 1 is in backwardation?

OPTIONS

A	Heating oil	✓ Correct	Correct: only heating oil has near > far.
B	Gold		Gold is flat (equal prices), not backwardation.
C	Coffee	✗ Your pick	Your pick: coffee rises 0.9850 → 1.0030 → 1.0290, which is contango.

Trigger “July | 1,274.8 | 0.9850 | 2.4105”
“December | 1,274.8 | 1.0290 | 2.1980”

Rule next time: Read each column from the nearest month (July) to the farthest (December). Heating oil falls 2.4105 → 2.3552 → 2.1980: near price higher than far = BACKWARDATION. Coffee rises = contango. Gold is flat.

FORMULA / RULE

Backwardation: F(near) > F(far) (downward-sloping futures curve)
Contango: F(near) < F(far)

WHY THE ANSWER IS A

1. Gold: 1,274.8 in all three months → flat.
2. Coffee: 0.9850 < 1.0030 < 1.0290 → contango.
3. Heating oil: 2.4105 > 2.3552 > 2.1980 → backwardation: **A**.

Apply to other questions: Do not use the coffee harvest story to decide; the exhibit alone answers 'which market is in backwardation'.

Mock 12 Q16 Accruals red flag → cash flow ratios (earnings quality), not DuPont or DCF

You **C** · Answer **A**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Worried that the balance sheet-based and cash flow-based accruals ratios (not shown) raise some concerns about the possible use of accruals to manage earnings, Lindgren asks Karin Holm, another analyst with the fund, for advice on what further type of analysis he should do as a follow-up on this issue.

Question 16. Holm's best answer to Lindgren's question about further analysis is that he should conduct a:

OPTIONS

A	cash flow ratio analysis.	✓ Correct	Correct: cash flow ratio analysis tests earnings quality.
B	DuPont analysis.		DuPont breaks ROE into margin, turnover and leverage; it does not detect earnings management.
C	discounted cash flow analysis.	✗ Your pick	Your pick: discounted cash flow analysis is a valuation tool, not a test of earnings quality.

Trigger “raise some concerns about the possible use of accruals to manage earnings”
“what further type of analysis he should do”

Rule next time: Concern that accruals are used to manage earnings → compare earnings with cash: cash flow ratios (e.g. operating cash flow before interest and taxes ÷ operating income). DuPont decomposes ROE; DCF values the company.

FORMULA / RULE

> Earnings quality check: cash flow from operations ÷ operating income (or net income)
> High accruals relative to cash flow → low-quality earnings

WHY THE ANSWER IS A

1. Accruals ratios show high accruals; the follow-up is to compare earnings with cash generated.
2. Cash flow ratio analysis does exactly that: **A**.

Apply to other questions: Match the tool to the goal: earnings quality → cash flow ratios; ROE drivers → DuPont; value → DCF.

Mock 12 Q18 **Multi-voting shares → principal–principal problem (controlling vs minority shareholders); voting caps do the opposite**

You **B** · Answer **A**

Corp

VIGNETTE (RELEVANT PART, UNEDITED)

The company is looking to expand its equity base to help fund a new workshop and support growth plans. Pryce, who is responsible for the company's financial management, is meeting with Anil Rao, a financial consultant, to explore Halden's equity financing options. Pryce asks if they were to go public, could they have a share structure similar to a company like Snap, where the co-founders could retain voting control of the company through the issuance of multi-voting ordinary (common) shares. Rao tells her that before Halden considers going public, there are other options available, and suggests they look for private equity investment. He mentions being familiar with a private equity fund that runs a socially responsible investment (SRI) pool that could potentially be interested in Halden.

Question 18. If Halden were to go public with the share structure similar to Snap that Pryce asked about, which governance issue would most likely arise?

OPTIONS

A	Principal–principal problem.	✓ Correct	Correct: principal–principal problem.
B	Voting cap restrictions.	✗ Your pick	Your pick: voting caps are restrictions on votes of large holders; multi-voting shares create the opposite.
C	Principal–agent problem.		Principal–agent arises when ownership and voting are dispersed (managers strong); the founders retain control here.

Trigger “retain voting control of the company through the issuance of multi-voting ordinary (common) shares” “co-founders”

Rule next time: Founders keep voting control with a small economic stake → conflict between controlling shareholders and minority shareholders = PRINCIPAL–PRINCIPAL. Principal–agent arises when ownership is dispersed. Voting caps are restrictions that LIMIT the votes of large holders.

FORMULA / RULE

> Concentrated control (multi-voting, family) → principal–principal
> Dispersed ownership, strong managers → principal–agent

WHY THE ANSWER IS A

1. Look at who controls the votes: the founders (controlling).
2. The conflict is with the minority shareholders: **A**.

Apply to other questions: Control concentrated → principal–principal; control dispersed → principal–agent.

Mock 12 Q26 **'Least likely violated': test each Standard against the facts (email to a SUBSET with a CHANGE in estimates; request NOT from the supervisor)**

You **A** · Answer **C**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Jane Pierce, having recently received her undergraduate degree in finance, accepted a job offer to join a large global brokerage firm as a research assistant. To manage training and oversight, all the firm's research assistants are supervised by a designated individual in the research department. Pierce was hired with the understanding she would register for the CFA Program exam within 90 days and be awarded her CFA designation within four years. Her first two days of work were spent in orientation, with a full day devoted to reviewing the firm's policies and procedures, including the firm's code of conduct. During the orientation, the chief compliance officer noted the firm claims compliance with the CFA Institute Code of Ethics and Standards of Professional Conduct. Shortly after Pierce is hired, she is accused of committing a potentially serious ethical violation. She is subsequently cleared of any wrongdoing.

Pierce worked hard getting to know the duties and responsibilities required in her new job. During her first month, the senior analyst she was assigned to work with asked her to distribute an email to a subset of asset management clients he meets with regularly. The email outlined his earnings model for a company under his coverage, his assumptions, the important factors surrounding his analysis, and the risk associated with those earnings. He also highlighted a change in his earnings expectations. Pierce was surprised by the request; in the past, she had emailed reports on behalf of other analysts, but those requests had always come from her supervisor, which was consistent with firm policy.

Question 26. Which CFA Institute Standard of Professional Conduct would least likely be violated if Pierce distributed the senior analyst's email as requested?

OPTIONS

A	Standard IV: Duties to Employers	✗ Your pick	Your pick: Standard IV is likely violated, because the request did not come from her supervisor as firm policy requires.
B	Standard III: Duties to Clients		Standard III is likely violated: a change in earnings expectations is sent only to a subset of clients.
C	Standard V: Investment Analysis, Recommendations, and Actions	✓ Correct	Correct: Standard V is not violated; the email explains the model, assumptions, risks and the change in expectations.

Trigger "least likely" (in the question)
"distribute an email to a subset of asset management clients"
"those requests had always come from her supervisor, which was consistent with firm policy"

Rule next time: Least likely violated = the Standard with NO problem. Selective email of a changed earnings view to a subset → Standard III (fair dealing); request bypassing the supervisor → firm policy / Standard IV; the email discloses assumptions and risks → Standard V is satisfied.

FORMULA / RULE

Standard III(B) Fair Dealing: disseminate changes in recommendations/estimates to all clients fairly
Standard IV(A) Loyalty: follow the firm's policies
Standard V(B) Communication: disclose process, assumptions, limitations and risks

WHY THE ANSWER IS C

1. Subset of clients + changed estimates → III(B).
2. Supervisor channel ignored → breach of firm policy (IV).
3. Assumptions and risks included → V satisfied: **C**.

Apply to other questions: 'Least likely' questions: list each option, mark the ones with a violation, choose the only one left clean.

Mock 12 Q29 Risk-neutral PD includes a premium for default timing uncertainty; the ACTUAL default rate does not

You **B** · Answer **A**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Gordon Ashby is the director of quantitative research at Westmere Asset Advisers, a fixed-income firm located in London. Rhea Vance recently joined Ashby's analyst team supporting credit analysis and credit risk modeling. After a brief orientation on the firm and its investment process, Ashby instructs Vance to develop a model that derives the credit valuation adjustment (CVA). Ashby makes the following comments regarding credit modeling.

Comment 1: The observed credit spread may also be higher than required for credit risk to compensate for a liquidity premium and tax considerations.

Comment 2: The factors to consider in modeling credit risk are the expected exposure to default loss, the risk-neutral probability of default, and the loss given default.

Comment 3: The use of an actual default rate in the model may overstate the observed value for a corporate bond because it includes a default risk premium associated with uncertainty over the timing of possible default loss.

Question 29. Ashby is least likely correct with regard to which point regarding credit modeling?

OPTIONS

A	Comment 3	✓ Correct	Correct: Comment 3 is the incorrect statement.
B	Comment 1	✗ Your pick	Your pick: Comment 1 is TRUE; the spread can include liquidity and tax effects.
C	Comment 2		Comment 2 is TRUE: expected exposure, risk-neutral PD and LGD are the three factors.

Trigger “least likely correct” (in the question)
“The use of an actual default rate in the model may overstate the observed value”
“it includes a default risk premium”

Rule next time: 'Least likely correct' = find the false comment. Comment 3 is false: the actual (real-world) default rate does NOT include the default risk premium; the risk-neutral PD does. Using the actual rate would overstate the bond value (understate the spread). Comments 1 and 2 are true.

FORMULA / RULE

Risk-neutral PD > actual PD (it includes the default risk premium)
CVA = Σ (expected exposure × risk-neutral PD × LGD, discounted)

WHY THE ANSWER IS A

1. Comment 1: spread above credit risk because of liquidity and tax: true.
2. Comment 2: the three CVA inputs: true.
3. Comment 3: actual rate lacks the premium: false → **A**.

Apply to other questions: Underline 'least likely' and check each comment against its rule; a true comment is NOT the answer.

Mock 12 Q30 **Risky bond valuation steps: VND − CVA; implied rates come from the PAR curve; recovery depends on seniority**

You **C** · Answer **C**

FI

Answered correctly; added at your request

VIGNETTE (RELEVANT PART, UNEDITED)

Exhibit 1: Steps in Valuing Risky Bonds, Arbitrage-Free Framework

Question 30. Which step in Exhibit 1 regarding valuing risky bonds has Vance most likely outlined correctly?

OPTIONS

A	Step 3		Step 3: recovery rate from the rating is wrong (seniority).
B	Step 2		Step 2: spot curve is wrong (par curve).
C	Step 1	✓ Correct	Correct: Step 1 is right.

Trigger “Using the spot curve, derive implied zero-coupon rates, discount factors, and forward rates” “Assign a recovery rate based on the company's credit rating”

Rule next time: Step 1 is correct (VND, CVA, fair value = VND − CVA). Step 2 is wrong: the tree is calibrated from the PAR curve, not the spot curve. Step 3 is wrong: the recovery rate reflects the SENIORITY of the debt (and asset nature), not the rating.

FORMULA / RULE

> Fair value = VND − CVA
> CVA = PV(LGD × risk-neutral PD × exposure)

WHY THE ANSWER IS C

1. Step 2 wrong: par curve.
2. Step 3 wrong: recovery by seniority.
3. Step 1 right: **C**.

Apply to other questions: Scan every bullet of a step; one wrong bullet makes the whole step wrong.

Mock 12 Q37 Which option is exercised? Rates UP → put exercised; rates DOWN → call exercised; conversion needs the stock UP

You **C** · Answer **A**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Paul Reeves, an investment adviser, meets with Helen Fischer to discuss her pending retirement and potential changes to her investment portfolio. Domestic economic activity has been weakening recently, and Reeves's outlook is that equity market values will be lower during the next year. He would like Fischer to consider reducing her equity exposure in favor of adding more fixed-income securities to the portfolio.

Government yields have remained low for an extended period, and Reeves suggests considering investment-grade corporate bonds to provide additional yield above government debt issues. Domestic inflation has accelerated, however, and the consensus forecast among economists is that the central bank, at its next meeting this month, will take actions that will lead to higher interest rates.

Exhibit 2: Selected Fixed-Rate Bonds of Orion Corporation

Bond	Annual Coupon	Type of Bond
Bond 1	1.6500%	Straight bond
Bond 2	1.6500%	Convertible bond: currently trading out of the money
Bond 3	1.6500%	Putable bond: putable at par one year and two years from now
Bond 4	1.6500%	Callable bond: callable at par without any protection periods

Question 37. Based on Exhibit 2, and assuming that the forecast for interest rates and Reeves's outlook for equity returns are validated, which bond's option is most likely to be exercised?

OPTIONS

A	Bond 3	✓ Correct	Correct: rates up → putable bond trades below par → put exercised.
B	Bond 2		Convertible: equities fall, so it stays out of the money.
C	Bond 4	✗ Your pick	Your pick: the issuer calls only when rates fall.

Trigger “actions that will lead to higher interest rates”
“equity market values will be lower”
“Putable bond: putable at par one year and two years from now”

Rule next time: The central bank will push rates up and equities will fall. Higher rates → bond price below par → the PUT (Bond 3) is exercised. The call (Bond 4) is exercised when rates fall; the convertible (Bond 2) needs a rising stock price.

FORMULA / RULE

> Rates ↑ → put exercised; rates ↓ → call exercised
> Stock ↑ → conversion exercised

WHY THE ANSWER IS A

1. Direction of rates: up.
2. Put at par is valuable when the bond price is below par: Bond 3: **A**.

Apply to other questions: Write the two forecasts (rates, equities) next to each option type before choosing.

Mock 12 Q39 Embedded option values and the yield curve: flatter/inverted → call value UP, put value DOWN (columns Call/Put, not Callable/Putable)

You **B** · Answer **C**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Reeves also states that he recently read a report issued by Hale and Company forecasting that the yield curve could invert within the next six months.

Exhibit 2: Selected Fixed-Rate Bonds of Orion Corporation

Bond	Annual Coupon	Type of Bond
Bond 1	1.6500%	Straight bond
Bond 2	1.6500%	Convertible bond: currently trading out of the money
Bond 3	1.6500%	Putable bond: putable at par one year and two years from now
Bond 4	1.6500%	Callable bond: callable at par without any protection periods

Question 39. If the Hale and Company forecast comes true, which of the following is most likely to occur? The value of the embedded option in:

OPTIONS

A	both Bond 3 and Bond 4 increases.		Wrong: put value falls, call value rises.
B	Bond 4 decreases.	✗ Your pick	Your pick: you read the Callable (bond price) column; the option itself (call) increases.
C	Bond 3 decreases.	✓ Correct	Correct: Bond 3's put option value decreases.

Trigger “forecasting that the yield curve could invert within the next six months”
“The value of the embedded option in”

Rule next time: Use your table 'Effect on option and bond values': Yield curve flatter (or inverted) → Call ↑, Callable ↓, Put ↓, Putable ↓. The question asks for the value of the embedded OPTION, so use the Call and Put columns: the call (Bond 4) rises, the put (Bond 3) falls.

FORMULA / RULE

> Table: flatter/inverted → Call ↑, Put ↓ (option values); Callable ↓, Putable ↓ (bond prices)

WHY THE ANSWER IS C

1. Question asks 'value of the embedded option': use the Call and Put columns.
2. Call ↑ (Bond 4), Put ↓ (Bond 3): **C**.

Apply to other questions: 'Value of the option' → Call/Put columns; 'value of the bond' → Callable/Putable columns. Underline the noun before you look at the table.

Mock 12 Q42 **Joint F-test: F < critical → FAIL to reject H0 (all slopes = 0) → none of the variables has explanatory power**

You **B** · Answer **C**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Analysis of Variance Table (ANOVA)

Source	Degrees of Freedom	Sum of Squares	Mean Square
Regression	4	214.00	53.50
Error	20	700.00	35.00
Total	24	914.00	

Table of the F-Distribution. Critical values for right-hand tail area equal to 0.05. Numerator: df1 and Denominator: df2

df2 \ df1	1	2	4	10	20
1	161.45	199.50	224.58	241.88	248.01
2	18.513	19.000	19.247	19.396	19.446
4	7.7086	6.9443	6.3882	5.9644	5.8025
10	4.9646	4.1028	3.4780	2.9782	2.7740
20	4.3512	3.4928	2.8661	2.3479	2.1242

Question 42. Marsh proceeds to test the hypothesis that none of the independent variables has significant explanatory power. Using the joint F-test for the significance of all slope coefficients, at a 5% level of significance:

OPTIONS

A	all of the independent variables have explanatory power.		All variables cannot be shown significant by the F-test; it only shows at least one.
B	at least one of the independent variables has explanatory power.	✗ Your pick	Your pick: 'at least one' is the conclusion only when F exceeds the critical value; here it does not.
C	none of the independent variables has explanatory power.	✓ Correct	Correct: fail to reject → none.

Trigger "none of the independent variables has significant explanatory power"
"Regression | 4 | 214.00 | 53.50"
"Error | 20 | 700.00 | 35.00"

Rule next time: H0: all slope coefficients = 0. F = MSR ÷ MSE = 53.50 ÷ 35.00 = 1.529 < critical 2.8661 (df 4 and 20) → fail to reject H0 → none of the independent variables has significant explanatory power. Reject would mean at least one has.

FORMULA / RULE

F = MSR ÷ MSE, df = k and n − k − 1
F < critical → fail to reject H0 (all slopes zero)

WHY THE ANSWER IS C

1. F = 53.50 ÷ 35.00 = 1.529.
2. Critical value (df1 = 4, df2 = 20): 2.8661.
3. 1.529 < 2.8661 → fail to reject H0: **C**.

Apply to other questions: Fail to reject = not enough evidence; joint F-test: reject → at least one matters; fail → none matters.